CONSTRUCTING THE KERNEL AND A COMPARISON TRANSFORMATION OVER QUASI-COMPACT SCHEMES WITH AFFINE DIAGONAL

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original equation 0FZC: sources/stacks-chow/tags.txt, source lines 19947–19947. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SG: sources/stacks-chow/tags.txt, source lines 2337–2337. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07TB: sources/stacks-chow/tags.txt, source lines 9931–9931. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZD: sources/stacks-chow/tags.txt, source lines 19948–19948. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZF: sources/stacks-chow/tags.txt, source lines 19950–19950. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XL: sources/stacks-chow/tags.txt, source lines 2522–2522. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let R be a ring. Let X and Y be schemes over R. Assume X is quasi-compact with affine diagonal. Let $F : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ be an R-linear, right exact functor which commutes with arbitrary direct sums. Then we can construct*

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(1) *a quasi-coherent module $\mathcal{K}$ on $X \times_R Y$, and*
(2) *a natural transformation $t : F \to F_{\mathcal{K}}$ where $F_{\mathcal{K}}$ denotes the functor (0FZC) such that $t : F \circ f_* \to F_{\mathcal{K}} \circ f_*$ is an isomorphism for every morphism $f : X' \to X$ whose source is an affine scheme.*

Proof. Consider a morphism $f' : X' \to X$ with X' affine. Since the diagonal of X is affine, we see that f' is an affine morphism (Morphisms, Lemma 01SG). Thus $f'_* : QCoh(\mathcal{O}_{X'}) \to QCoh(\mathcal{O}_X)$ is an R-linear exact functor (Cohomology of Schemes, Lemma 01XC) which commutes with direct sums (Cohomology of Schemes, Lemma 07TB). Thus $F \circ f'_*$ is an R-linear, right exact functor which commutes with arbitrary direct sums. Whence $F \circ f'_* = F_{\mathcal{K}'}$ for some $\mathcal{K}'$ on $X' \times_R Y$ by Lemma 0FZD. Moreover, given a morphism $f'' : X'' \to X'$ with X'' affine we obtain a canonical identification $(f'' \times \mathrm{id}_Y)^*\mathcal{K}' = \mathcal{K}''$ by the references already given combined with Lemma 0FZF. These identifications satisfy a cocycle condition given another morphism $f''' : X''' \to X''$ which we leave it to the reader to spell out.

Choose an affine open covering $X = \bigcup_{i=1,\ldots,n} U_i$. Since the diagonal of X is affine, we see that the intersections $U_{i_0 \ldots i_p} = U_{i_0} \cap \ldots \cap U_{i_p}$ are affine. As above the inclusion morphisms $j_{i_0 \ldots i_p} : U_{i_0 \ldots i_p} \to X$ are affine. Denote $\mathcal{K}_{i_0 \ldots i_p}$ the quasi-coherent module on $U_{i_0 \ldots i_p} \times_R Y$ corresponding to $F \circ j_{i_0 \ldots i_p *}$ as above. By the above we obtain identifications

$$\mathcal{K}_{i_0 \ldots i_p} = \mathcal{K}_{i_0 \ldots \hat{i}_j \ldots i_p}|_{U_{i_0 \ldots i_p} \times_R Y}$$

which satisfy the usual compatibilites for glueing. In other words, we obtain a unique quasi-coherent module $\mathcal{K}$ on $X \times_R Y$ whose restriction to $U_{i_0 \ldots i_p} \times_R Y$ is $\mathcal{K}_{i_0 \ldots i_p}$ compatible with the displayed identifications.

Next, we construct the transformation t. Given a quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ denote $\mathcal{F}_{i_0 \ldots i_p}$ the restriction of $\mathcal{F}$ to $U_{i_0 \ldots i_p}$ and denote $(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0 \ldots i_p}$ the restriction of $\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K}$ to $U_{i_0 \ldots i_p} \times_R Y$. Observe that

$$\begin{aligned} F(j_{i_0 \ldots i_p *}\mathcal{F}_{i_0 \ldots i_p}) &= \mathrm{pr}_{i_0 \ldots i_p, 2, *}(\mathrm{pr}_{i_0 \ldots i_p, 1}^*\mathcal{F}_{i_0 \ldots i_p} \otimes \mathcal{K}_{i_0 \ldots i_p}) \\ &= \mathrm{pr}_{i_0 \ldots i_p, 2, *}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0 \ldots i_p} \end{aligned}$$

where $\mathrm{pr}_{i_0 \ldots i_p, 2} : U_{i_0 \ldots i_p} \times_R Y \to Y$ is the projection and similarly for the other projection. Moreover, these identifications are compatible with the displayed identifications in the previous paragraph. Recall, from Cohomology of Schemes, Lemma 01XL that the relative Čech complex

$$\bigoplus \mathrm{pr}_{i_0, 2, *}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0} \to \bigoplus \mathrm{pr}_{i_0 i_1, 2, *}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0 i_1} \to \bigoplus \mathrm{pr}_{i_0 i_1 i_2, 2, *}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0 i_1 i_2} \to \ldots$$

computes $R\mathrm{pr}_{2,*}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})$. Hence the cohomology sheaf in degree 0 is $F_{\mathcal{K}}(\mathcal{F})$. Thus we obtain the desired map $t : F(\mathcal{F}) \to F_{\mathcal{K}}(\mathcal{F})$ by contemplating the following commutative diagram

$$\begin{array}{ccccc} & & F(\mathcal{F}) & \longrightarrow & \bigoplus F(j_{i_0 *}\mathcal{F}_{i_0}) & \longrightarrow & \bigoplus F(j_{i_0 i_1 *}\mathcal{F}_{i_0 i_1}) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F_{\mathcal{K}}(\mathcal{F}) & \longrightarrow & \bigoplus \mathrm{pr}_{i_0, 2, *}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0} & \longrightarrow & \bigoplus \mathrm{pr}_{i_0 i_1, 2, *}(\mathrm{pr}_1^*\mathcal{F} \otimes \mathcal{K})_{i_0 i_1} \end{array}$$

We obtain the top row by applying F to the (exact) complex $0 \to \mathcal{F} \to \bigoplus j_{i_0 *}\mathcal{F}_{i_0} \to \bigoplus j_{i_0 i_1 *}\mathcal{F}_{i_0 i_1}$ (but since F is not exact, the top row is just a complex and not necessarily exact). The solid vertical arrows are the identifications above. This does indeed define the dotted arrow as desired. The arrow is functorial in $\mathcal{F}$; we omit the details.

We still have to prove the final assertion. Let $f : X' \to X$ be as in the statement of the lemma and let $\mathcal{K}'$ be the quasi-coherent module on $X' \times_R Y$ constructed in the first paragraph of the proof. If the morphism $f : X' \to X$ maps into one of the opens U_i, then the result follows from Lemma 0FZF because in this case we know that $\mathcal{K}_i = \mathcal{K}|_{U_i \times_R Y}$ pulls back to $\mathcal{K}$. In general, we obtain an affine open covering $X' = \bigcup U'_i$ with $U'_i = f^{-1}(U_i)$ and we obtain isomorphisms $\mathcal{K}'|_{U'_i} = f_i^* \mathcal{K}_i$ where $f_i : U'_i \to U_i$ is the induced morphism. These morphisms satisfy the compatibility conditions needed to glue to an isomorphism $\mathcal{K}' = f^*\mathcal{K}$ and we conclude. Some details omitted. □